



Mid-range capacity incubation without compromise

Thermo Scientific Cytomat 2 C-LiN automated incubators and storage devices

The next generation of advanced incubation and storage solutions for cell culture applications:

- Dependable sample protection
- Wide temperature range
- Super-fast access
- TRUE orbital shaking

Scientists need to trust their incubator to provide the right environment for their cells. This includes maintaining consistent temperatures, humidity conditions, gas environments, and shaking patterns.

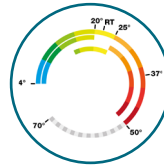
Providing ideal solutions for mid-capacity incubation and storage, Thermo Scientific™ Cytomat™ 2 incubators combine superior environmental conditions with super-fast access and reliable sample protection.

- Wide temperature range available, from 4 to 70°C
- Fully automated decontamination routine (ContraCon)
- Controlled humidity with external water tank
- Reliable, smooth and precise handling



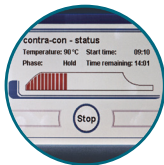
Unrivalled speed

The fastest incubators in their class, with a mean access time of 12 seconds. Individual speed settings per plate and adjustable incremental speed settings are available.



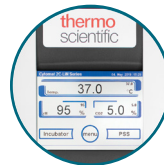
Expanded temperature range

Incubators and storage solutions covering a range of temperatures from 4 to 70°C, with the fastest access in their class.



Assured decontamination

Cytomat 2 incubators have a fully automated ContraCon thermal decontamination routine, well established in tens of thousands of Thermo Scientific™ Heracell™ and other Cytomat incubators throughout the world.



Easy everyday use

5-inch LED glove-friendly display with intuitive operation. Includes read-out messaging, warning alerts, and inventory data.



Flexible solution

Suitable for benchtop and under the bench, with multiple loading/unloading location options.

Flexibility for all solutions

Options and accessories

Wide range of transfer stations



Standard transfer station

Supplies the microplate in portrait format. A 90° turnable version is also available for presentation in landscape format.



Swap station (180°)

180° turnable swap station to accelerate your handling process.



Linear transfer station

Available in different lengths (300, 400 and 500 mm) for extended reach. Delivers the microplate in portrait format with further options for landscape presentation or lateral movement.



Plate lift for under-bench solution

Stepper motor driven (SMD) technology offers microplate movement to a predefined height within a 500 mm stroke.

Options for your precise requirements



Precise humidity control

HydraSmart ultrasonic vaporizer technology for controlled humidity and an external water reservoir for maximum convenience and minimal contamination risk.



Solid copper antimicrobial material

Only solid pure antimicrobial copper chambers are proven to be effective against contamination.



TRUE Orbital Shaking Control

Produces ideal growth conditions for cells requiring consistent TRUE orbital shaking.

Further general options

- 1D barcode reader
- P3-plate detection in stacker
- Various stacker pitch sizes for different plate heights
- Customized stackers for different labware
- HydraSmart plus: Humidity control $\leq 10\%$ rH for dry environment

Specific options

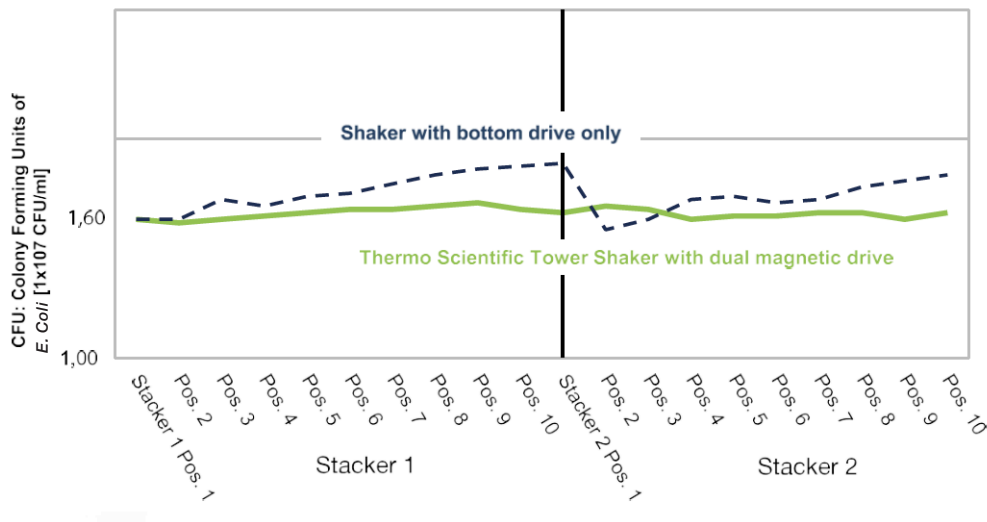
- O₂ control (1–21% or 5–90%)
- hH: high humidity up to 98% rH at 37°C*
- Door Left hinges (DL)
- H₂O₂ disinfection ports (GDF)
- Air Purging System (APS): CO₂ reduction during cell growth down to 1%

* Condensate formation is possible

TRUE orbital shaking

The Cytomat 2 automated incubator with tower shaker is the ultimate solution for TRUE orbital shaking. Perfect for applications that demand precise sample agitation and cells in suspension, this innovative incubator sets a new standard in automated shaking incubation.

The synchronized dual magnetic drive system, positioned at both the top and bottom, guarantees a uniform shaking diameter across all microplates, regardless of load, position, or number of plates.



Key Applications

- Cell culture (mammalian)
- Microbial culture
- Genomics
- Biologics
- Luminex (ELISA, biomarkers)
- Transient transfection

Key Applications

- Drug discovery
- Biopharmaceuticals
- Life sciences research
- Diagnostics

Bacterial growth consistency over the entire two stackers (bottom to top) of the tower shaking option

The graph shows optimum bacterial growth with the dual magnetic drive system across all plates (green line; standard deviation: 2.7%). In comparison, the technology with only bottom drive shaking reveals inconsistent actual growth (standard deviation: 10.1%)

Tower shaker features

1. Active microplate clamp mechanism secures plate and lid separately; no abrasions of the microplates occur during shaking
2. Inductive wear-free driving mechanism allows for no particle emission or belts to change
3. Superior shaking technology with individual speed settings of each tower shaker from 300 to 1200 rpm with a shaking diameter of 3 mm (2 mm option available) offers a wide range of application setups
4. Adjacent tower shaker continues to shake upon plate removal for increased repeatability of the assays, even with unbalanced load
5. Mix and match standard and tower shaker stacker for maximum flexibility for assay adoption

One product series for every application

Developed by the inventors of microbiological incubators, and taking advantage of over two decades of unequalled automation experience, the Cytomat 2 automated incubator offers the broadest set of capabilities for incubation and storage. With minimal effort it delivers robust and consistent environmental conditions, the fastest access time in its class, and TRUE orbital shaking.

The Cytomat 2 automated incubator is the solution for all applications—from random ambient through low temperature and room incubation to 37°C incubation and hybridization, and beyond.

Model	Controlled temperature range	Low-temperature storage	Ambient random access storage	Room temperature incubation	Biochemical assays	Cell assays	Mammalian and microbial cell incubation	Short-term incubation	Tower shaker control
Cytomat 2 C	(Ambient + 10°C) - 50°C				●	●	●	●	
Cytomat 2 C425	4 - 25°C	●		●	●			●	●
Cytomat 2 C450	4 - 50°C	●	●	●	●	●	●	●	●
Cytomat 2 C470	4 - 70°C	●	●	●	●	●	●	●	●

●

 Capability

●

 Good capability

Overview of the tower shaker stacker

(300–1200 rpm; max. load of 850 g per tower shaker at 1200 rpm)

Accessory	Description
Tower shaker (pitch 23)	Capacity: 16 plates Plate height: ≤ 18 mm 3mm shaking diameter (optional 2mm)
Tower shaker (pitch 27)	Capacity: 13 plates Plate height: ≤ 22 mm 3 mm shaking diameter (optional 2mm)
Tower shaker (pitch 52)	Capacity: 7 plates Plate height: ≤ 47 mm 3 mm shaking diameter (optional 2mm)
Tower shaker stacker now available with robot-friendly lids by Enzymscreen.	
Tower shaker (3mm, pitch 60)	Capacity: 6 plates Plate height: ≤ 55 mm 3 mm shaking diameter Note: For deep-well plates with robot-friendly lids (by Enzymscreen)

Cytomat 2 automated incubators and storage devices

Discover your advanced incubation and storage solution today

- **Wide temperature ranges available**
- **Mean access time below 12 seconds**
- **Automated decontamination routine with ContraCon**
- **Controlled humidity through HydraSmart technology**
- **TRUE orbital shaking control**



 Learn more at thermofisher.com/cytomat

thermo scientific